Load Short Form
Entire House

Job: 2325 Sq. Ft.
Date: 7/14/2026
By: Steven Ferrell

Project Information

For: Florence, Infinity Homes
Ocala, FL

Design Information

	Htg	Clg
Outside db (°F)	34	91
Inside db (°F)	70	75
Design TD (°F)	36	16
Daily range	-	M
Inside humidity (%)	50	50
Moisture difference (gr/lb)	32	44

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

HEATING EQUIPMENT

Make	Carrier
Trade	15 SEER2 HP
Model	GH5SAN54200AA0
AHRI ref	214101864

Efficiency	7.5 HSPF2	
Heating input		
Heating output	39500	Btuh @ 47°F
Temperature rise	29	°F
Actual air flow	1240	cfm
Air flow factor	0.032	cfm/Btuh
Static pressure	0.50	in H2O
Space thermostat		

Capacity balance point = 36 °F

Backup:
Input = 8 kW, Output = 25611 Btuh, 100 AFUE

COOLING EQUIPMENT

Make	Carrier
Trade	15 SEER2 HP
Cond	GH5SAN54200AA0
Coil	FJ5ANBC42L00
AHRI ref	214101864

Efficiency	12.0 EER2,14.5 SEER2	
Sensible cooling	28700	Btuh
Latent cooling	12300	Btuh
Total cooling	41000	Btuh
Actual air flow	1240	cfm
Air flow factor	0.044	cfm/Btuh
Static pressure	0.50	in H2O
Load sensible heat ratio	0.79	

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
Kitchen	214	1678	2761	53	122
Laundry	53	122	581	4	26
Foyer/Dining	371	4853	4135	153	183
Bath #3	53	856	272	27	12
Bedroom #3	159	2020	937	64	42
Bedroom #2	141	1803	871	57	39
M. Bath	118	1563	516	49	23
Bedroom #4	155	4948	2283	156	101
WC	21	49	28	2	1
Gathering	318	3992	6795	126	301
Bath #2	60	959	302	30	13
WIC 3	20	46	26	1	1
WIC 4	42	1060	287	33	13
Master Bedroom	234	5864	3537	185	157

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.

Study/Bed #5	141	3110	1078	98	48
Flex	214	6364	3572	201	158
Entire House d	2325	39287	27982	1240	1240
Other equip loads		2684	1223		
Equip. @ 0.96 RSM			28153		
Latent cooling			7629		
TOTALS	2325	41971	35782	1240	1240

Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.

Building Analysis
Entire House

Job: 2325 Sq. Ft.
Date: 7/14/2026
By: Steven Ferrell

Project Information

For: Florence, Infinity Homes
Ocala, FL

Design Conditions

Location:
Ocala, FL, US
Elevation: 87 ft
Latitude: 29°N

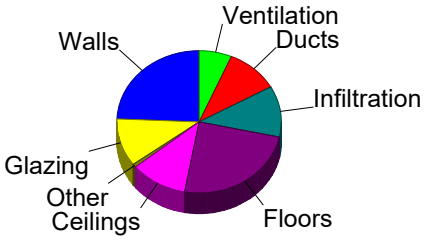
Outdoor:	**Heating**	**Cooling**
Dry bulb (°F)	34	91
Daily range (°F)	-	17 (M)
Wet bulb (°F)	-	76
Wind speed (mph)	15.0	7.5

Indoor:	**Heating**	**Cooling**
Indoor temperature (°F)	70	75
Design TD (°F)	36	16
Relative humidity (%)	50	50
Moisture difference (gr/lb)	31.8	44.4

Infiltration:	
Method	Simplified
Construction quality	Average
Fireplaces	0

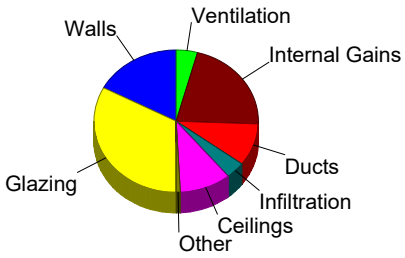
Heating

Component	Btuh/ft²	Btuh	% of load
Walls	4.9	10209	24.3
Glazing	14.7	4564	10.9
Doors	14.0	295	0.7
Ceilings	2.0	4719	11.2
Floors	4.4	10252	24.4
Infiltration	2.3	4875	11.6
Ducts		4373	10.4
Piping		0	0
Humidification		0	0
Ventilation		2684	6.4
Adjustments		0	
Total		**41971**	**100.0**



Cooling

Component	Btuh/ft²	Btuh	% of load
Walls	2.4	5028	17.2
Glazing	30.7	9567	32.8
Doors	11.4	240	0.8
Ceilings	1.3	2953	10.1
Floors	0	0	0
Infiltration	0.5	1110	3.8
Ducts		2824	9.7
Ventilation		1223	4.2
Internal gains		6260	21.4
Blower		0	0
Adjustments		0	
Total		**29204**	**100.0**



Latent Cooling Load = 7629 Btuh
Overall U-value = 0.234 Btuh/ft²-°F, Window / Floor Area = 13.5 %

Data entries checked.

Component Constructions
Entire House

Job: 2325 Sq. Ft.
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By: Steven Ferrell

Project Information

For: Florence, Infinity Homes
Ocala, FL

Design Conditions

Location:
Ocala, FL, US
Elevation: 87 ft
Latitude: 29°N

Outdoor:	**Heating**	**Cooling**
Dry bulb (°F)	34	91
Daily range (°F)	-	17 (M)
Wet bulb (°F)	-	76
Wind speed (mph)	15.0	7.5

Indoor:	**Heating**	**Cooling**
Indoor temperature (°F)	70	75
Design TD (°F)	36	16
Relative humidity (%)	50	50
Moisture difference (gr/lb)	31.8	44.4

Infiltration:

Method	Simplified
Construction quality	Average
Fireplaces	0

Construction descriptions	Area ft²	U-value Btuh/ft²-°F	Insul R ft²-°F/Btuh	Htg HTM Btuh/ft²	Loss Btuh	Clg HTM Btuh/ft²	Gain Btuh
Walls							
Blk wall, stucco ext, r-4 ext bd ins, 8" thk, 1/2" gypsum board int fnsh	655	0.143	4.1	5.15	3374	2.58	1692
	226	0.143	4.1	5.15	1165	2.58	584
	556	0.143	4.1	5.15	2864	2.58	1436
	374	0.143	4.1	5.15	1924	2.58	965
	1812	0.143	4.1	5.15	9327	2.58	4677
Partitions							
Frm wall, r-13 cav ins, 1/2" gypsum board int fnsh, 2"x4" wood frm, 16" o.c. stud	269	0.091	13.0	3.28	881	1.31	351
Windows							
2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; 1.33 ft overhang (1.33 ft window ht, 1.17 ft sep.); 6.67 ft head ht	8	0.340	0	12.2	98	10.6	85
	8	0.340	0	12.2	98	10.6	85
	16	0.340	0	12.2	196	10.6	170
2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; 1.33 ft overhang (5 ft window ht, 1.17 ft sep.); 6.67 ft head ht	30	0.340	0	12.2	367	10.6	319
	15	0.340	0	12.2	184	10.6	160
	45	0.340	0	12.2	551	32.4	1456
	90	0.340	0	12.2	1102	21.5	1935
2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; 1.33 ft overhang (6 ft window ht, 1.17 ft sep.); 6.67 ft head ht	36	0.340	0	12.2	441	32.4	1165
2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 6.33 ft overhang (6 ft window ht, 3.25 ft sep.); 6.67 ft head ht	27	0.340	0	12.2	330	35.1	948

2 glazing, clr low-e outr, argon gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, argon gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 6.33 ft overhang (8 ft window ht, 3.25 ft sep.); 8 ft head ht	24	0.340	0	12.2	294	36.9	886
2 glazing, clr low-e outr, argon gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, argon gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 16.67 ft overhang (8 ft window ht, 3.25 ft sep.); 8 ft head ht	24	0.340	0	12.2	294	20.6	495
2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk: 2 glazing, clr low-e outr, air gas, vnl frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 9 ft overhang (5 ft window ht, 0.67 ft sep.); 6.67 ft head ht	22	0.340	0	12.2	275	20.6	464
4A5-2omd: 2 glazing, clr low-e outr, air gas, mtl no brk frm mat, clr innr, 1/4" gap, 1/4" thk; NFRC rated (SHGC=0.32); 50% blinds 45°, medium; foreground = new concrete (0.32); 9 ft overhang (8 ft window ht, 0.67 ft sep.); 8 ft head ht	72	0.630	0	22.7	1633	28.7	2063
Doors							
11D0: Door, wd sc type	21	0.390	0	14.0	295	11.4	240
Ceilings							
Attic ceiling, asphalt shingles roof mat, r-20 roof ins, 1/2" gypsum board int fnsh	2316	0.408	38.0	2.04	4719	1.28	2953
Floors							
22A-tpm: Bg floor, heavy dry or light damp soil, on grade depth, carp 80% flr fnsh	241	1.180	0	42.5	10252	0	0

Duct System Summary
Entire House

Job: 2325 Sq. Ft.
Date: 7/14/2026
By: Steven Ferrell

Project Information

For: Florence, Infinity Homes
Ocala, FL

	Heating	Cooling
External static pressure	0.50 in H2O	0.50 in H2O
Pressure losses	0.31 in H2O	0.31 in H2O
Available static pressure	0.19 in H2O	0.19 in H2O
Supply / return available pressure	0.115 / 0.075 in H2O	0.115 / 0.075 in H2O
Lowest friction rate	0.061 in/100ft	0.061 in/100ft
Actual air flow	1240 cfm	1240 cfm
Total effective length (TEL)	312 ft	

Supply Branch Detail Table

Name		Design (Btuh)	Htg (cfm)	Clg (cfm)	Design FR	Diam (in)	H x W (in)	Duct Matl	Actual Ln (ft)	Ftg.Eqv Ln (ft)	Trunk
Bath #2	h	959	30	13	0.067	4.0	0x 0	VlFx	36.3	135.0	st3
Bath #3	h	856	27	12	0.063	4.0	0x 0	VlFx	48.3	135.0	st1
Bedroom #2	h	1803	57	39	0.068	***5.0***	***0x 0***	VlFx	39.3	130.0	st3
Bedroom #3	h	2020	64	42	0.063	***5.0***	***0x 0***	VlFx	46.5	135.0	st1
Bedroom #4	h	4948	156	101	0.061	7.0	0x 0	VlFx	53.9	135.0	st1
Flex	h	6364	201	158	0.065	8.0	0x 0	VlFx	40.8	135.0	st3
Foyer/Dining	c	4135	153	183	0.072	8.0	0x 0	VlFx	30.5	130.0	st3
Gathering	c	3398	63	151	0.064	7.0	0x 0	VlFx	45.7	135.0	st1
Gathering-A	c	3398	63	151	0.063	7.0	0x 0	VlFx	47.0	135.0	st1
Kitchen-A	c	2761	53	122	0.076	6.0	0x 0	VlFx	26.9	125.0	st2
Laundry	c	581	4	26	0.071	4.0	0x 0	VlFx	31.9	130.0	st2
M. Bath	h	1563	49	23	0.080	4.0	0x 0	VlFx	23.3	120.0	st2
Master Bedroom	h	5864	185	157	0.070	8.0	0x 0	VlFx	30.4	135.0	st2
Study/Bed #5	h	3110	98	48	0.078	5.0	0x 0	VlFx	27.5	120.0	st2
WC	h	49	2	1	0.079	4.0	0x 0	VlFx	20.8	125.0	st2
WIC 3	h	46	1	1	0.076	4.0	0x 0	VlFx	26.8	125.0	st2
WIC 4	h	1060	33	13	0.076	4.0	0x 0	VlFx	26.3	125.0	st2

Supply Trunk Detail Table

Name	Trunk Type	Htg (cfm)	Clg (cfm)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Duct Material	Trunk
st1	Peak AVF	373	456	0.061	580	12.0	0 x 0	VinlFlx	
st2	Peak AVF	426	391	0.070	542	12.0	0 x 0	VinlFlx	
st3	Peak AVF	441	394	0.065	562	12.0	0 x 0	VinlFlx	

Bold/italic values have been manually overridden

Return Branch Detail Table

Name	Grille Size (in)	Htg (cfm)	Clg (cfm)	TEL (ft)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Stud/Joist Opening (in)	Duct Matl	Trunk
rb1	0x 0	231	172	118.5	0.063	662	8.0	0x 0		VlFx	rt1
rsb7	0x 0	57	39	116.2	0.064	417	***5.0***	***0***x ***0***		VlFx	rt1
rb10	0x 0	98	48	77.3	0.097	720	5.0	0x 0		VlFx	
rb2	18x 25	854	982	122.8	0.061	556	18.0	0x 0		VlFx	rt1

Return Trunk Detail Table

Name	Trunk Type	Htg (cfm)	Clg (cfm)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Duct Material	Trunk
rt1	Peak AVF	1142	1192	0.061	546	20.0	0 x 0	VinlFlx	

Bold/italic values have been manually overridden

Manual S Compliance Report
Entire House

Job: 2325 Sq. Ft.
Date: 7/14/2026
By: Steven Ferrell

Project Information

For: Florence, Infinity Homes
Ocala, FL

Cooling Equipment

Design Conditions

Outdoor design DB:	91.4°F	Sensible gain:	29204	Btuh	Entering coil DB:	76.0°F
Outdoor design WB:	75.7°F	Latent gain:	7629	Btuh	Entering coil WB:	63.6°F
Indoor design DB:	75.0°F	Total gain:	36834	Btuh		
Indoor RH:	50%	Estimated airflow:	1240	cfm		

Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Split ASHP				
Manufacturer:	Carrier		Model:	GH5SAN54200AA0+FJ5ANBC42L00	
Actual airflow:	1240	cfm			
Sensible capacity:	31245	Btuh	107% of load		
Latent capacity:	8271	Btuh	108% of load		
Total capacity:	39516	Btuh	107% of load	SHR:	79%

Heating Equipment

Design Conditions

Outdoor design DB:	34.0°F	Heat loss:	41971	Btuh	Entering coil DB:	67.8°F
Indoor design DB:	70.0°F					

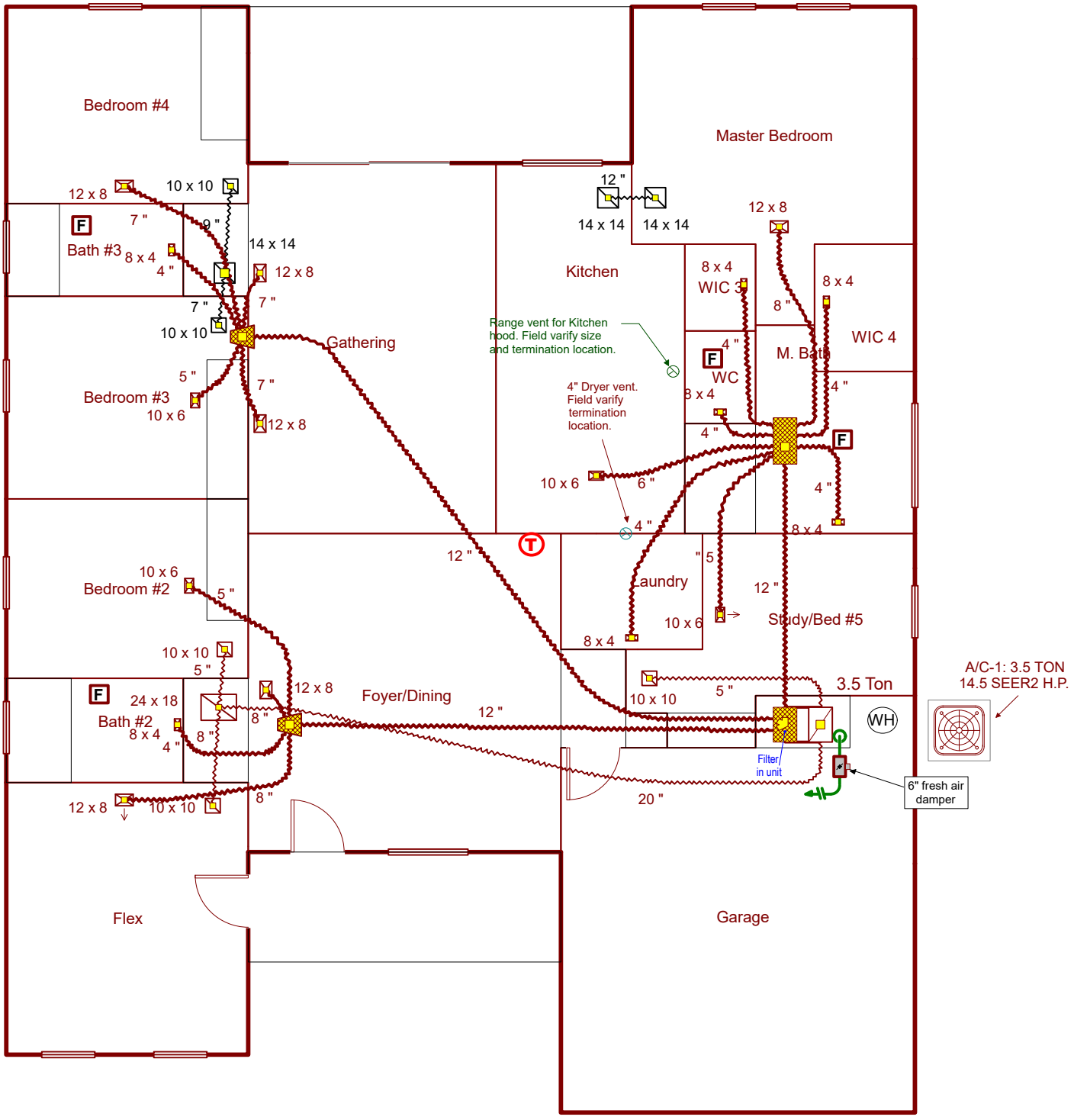
Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Split ASHP						
Manufacturer:	Carrier			Model:	GH5SAN54200AA0+FJ5ANBC42L00		
Actual airflow:	1240	cfm					
Output capacity:	32487	Btuh	77% of load			Capacity balance:	36 °F
Supplemental heat required:	9485	Btuh				Economic balance:	-99 °F

Backup equipment type:	Elec strip				
Manufacturer:			Model:		
Actual airflow:	1240	cfm			
Output capacity:	7.5	kW	61% of load	Temp. rise:	41 °F

Meets all requirements of ACCA Manual S.

Level 1



Job #: 2325 Sq. Ft.
Performed by Steven Ferrell for:
Florence

Ocala, FL

Scale: 1 : 101

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